



FACULTÉ DES SCIENCES ET DES TECHNOLOGIES(FST)

TD N°5_ – ADMIN-SYSTEME

COURS: ADMINISTRATION DES SYSTÈMES
INFORMATIQUES

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L'objectif de ce TD est de:

1. sécuriser un serveur Linux
2. protéger les accès SSH
3. filtrer le trafic réseau
4. bloquer les attaques automatisées
5. analyser la sécurité d'un système
6. installer un outil d'audit de sécurité
7. interpréter un rapport d'audit
8. identifier les failles potentielles
9. proposer des améliorations de sécurité

Analyse initiale du serveur

Vérifier les ports ouverts

```
valid_lft forever preferred_lft forever
lens@lens:~$ ss -tuln
Netid      State      Recv-Q     Send-Q      Local Address:Port
udp        UNCONN     0           0            127.0.0.54:53
udp        UNCONN     0           0            127.0.0.53%lo:53
udp        UNCONN     0           0            192.168.43.79%enp0s3:68
udp        UNCONN     0           0            0.0.0.0:111
udp        UNCONN     0           0            127.0.0.1:161
udp        UNCONN     0           0            [::]:111
udp        UNCONN     0           0            [::1]:161
tcp        LISTEN     0           4096         127.0.0.54:53
tcp        LISTEN     0           4096         0.0.0.0:22
tcp        LISTEN     0           100         0.0.0.0:25
tcp        LISTEN     0           511         0.0.0.0:80
tcp        LISTEN     0           4096         0.0.0.0:111
tcp        LISTEN     0           4096         127.0.0.53%lo:53
tcp        LISTEN     0           4096         0.0.0.0:10050
tcp        LISTEN     0           4096         [::]:22
tcp        LISTEN     0           100         [::]:25
tcp        LISTEN     0           4096         [::]:111
tcp        LISTEN     0           4096         [::]:10050
lens@lens:~$
```

Vérifier les services actifs

UNIT	LOAD	ACTIVE	SUB	DESCR
apparmor.service	loaded	active	exited	Load A
apport.service	loaded	active	exited	automa
blk-availability.service	loaded	active	exited	Availa
console-setup.service	loaded	active	exited	Set co
cron.service	loaded	active	running	Regula
dbus.service	loaded	active	running	D-Bus
fail2ban.service	loaded	active	running	Fail2B
finalrd.service	loaded	active	exited	Create
getty@tty1.service	loaded	active	running	Getty
keyboard-setup.service	loaded	active	exited	Set th
kmod-static-nodes.service	loaded	active	exited	Create
lvm2-monitor.service	loaded	active	exited	Monito
ModemManager.service	loaded	active	running	Modem
multipathd.service	loaded	active	running	Device
networkd-dispatcher.service	loaded	active	running	Dispat
nftables.service	loaded	active	exited	nftabl
nginx.service	loaded	active	running	A high
php8.3-fpm.service	loaded	active	running	The Ph
plymouth-quit-wait.service	loaded	active	exited	Hold u
plymouth-quit.service	loaded	active	exited	Termin
plymouth-read-write.service	loaded	active	exited	Tell F
polkit.service	loaded	active	running	Author
postfix.service	loaded	active	exited	Postfi
postfix@-.service	loaded	active	running	Postfi
rpcbind.service	loaded	active	running	RPC b1
rsyslog.service	loaded	active	running	System
setvtrngb.service	loaded	active	exited	Set co
snapped.service	loaded	active	exited	Load A
snapped.seeded.service	loaded	active	exited	Wait u
snmpd.service	loaded	active	running	Simple
ssh.service	loaded	active	running	OpenBS
sysstat.service	loaded	active	exited	Resets
systemd-binfmt.service	loaded	active	exited	Set Up
systemd-fsck@dev-disk-by\x2duuid-ddf25e8d\x2d86fd\x2d4788\x2d9212\x2d8c827860e133.service	loaded	active	exited	File S
systemd-journal-flush.service	loaded	active	exited	Flush
systemd-journald.service	loaded	active	running	Journa
systemd-logind.service	loaded	active	running	User L
systemd-modules-load.service	loaded	active	exited	Load K
systemd-networkd-wait-online.service	loaded	active	exited	Wait f
systemd-networkd.service	loaded	active	running	Networ
systemd-random-seed.service	loaded	active	exited	Load/S
systemd-remount-fs.service	loaded	active	exited	Remoun
systemd-resolved.service	loaded	active	running	Networ
systemd-sysctl.service	loaded	active	exited	Apply
systemd-timesyncd.service	loaded	active	running	Networ
systemd-tmpfiles-setup-dev-early.service	loaded	active	exited	Create
systemd-tmpfiles-setup-dev.service	loaded	active	exited	Create
systemd-tmpfiles-setup.service	loaded	active	exited	Create

lines 1-49

Vérifier les utilisateurs

```

lens@lens:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/usr/sbin/nologin
systemd-timesync:x:997:997:systemd Time Synchronization:/:/usr/sbin/nologin
dhcpcd:x:100:65534:DHCP Client Daemon,,,:/usr/lib/dhcpcd:/bin/false
messagebus:x:101:102::/nonexistent:/usr/sbin/nologin
systemd-resolve:x:992:992:systemd Resolver:/:/usr/sbin/nologin
pollinate:x:102:1::/var/cache/pollinate:/bin/false
polkitd:x:991:991:User for polkitd:/:/usr/sbin/nologin
syslog:x:103:104::/nonexistent:/usr/sbin/nologin
uuidd:x:104:105::/run/uuidd:/usr/sbin/nologin
tcpdump:x:105:107::/nonexistent:/usr/sbin/nologin
tss:x:106:108:TPM software stack,,,:/var/lib/tpm:/bin/false
landscape:x:107:109::/var/lib/landscape:/usr/sbin/nologin
fwupd-refresh:x:989:989:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
usbmux:x:108:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
lens:x:1000:1000:Sandro:/home/lens:/bin/bash
sshd:x:109:65534::/run/sshd:/usr/sbin/nologin
backupuser:x:1001:1001:sandro lens,5343,353786888,6363738822,0000:/home/backupuser:/bin/bash
backupusererr:x:1002:1002:sandro lens,5566566556,566676677,566676666,0000:/home/backupusererr:/bin/bash
_rpc:x:110:65534::/run/rpcbind:/usr/sbin/nologin
postfix:x:111:111::/var/spool/postfix:/usr/sbin/nologin
nagios:x:112:113::/var/lib/nagios:/usr/sbin/nologin
zabbix:x:113:114::/var/lib/zabbix:/usr/sbin/nologin
Debian-snmp:x:114:115::/var/lib/snmp:/bin/false
lens@lens:~$

```

Sécurisation de SSH

```
# ListenAddress.
#
# For changes to take effect, run:
#
#   systemctl daemon-reload
#   systemctl restart ssh.socket
#
Port 2222
#AddressFamily any
#ListenAddress 10.79.242.79
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin no
#StrictModes yes
MaxAuthTries 3
#MaxSessions 10
#PubkeyAuthentication no
PasswordAuthentication no

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2

#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody
AllowUsers admin

lens@lens:~$ sudo systemctl restart ssh
```

Configuration du Firewall

Installation de ufw

```
lens@lens:~$ sudo apt install ufw -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  ufw
0 upgraded, 1 newly installed, 0 to remove and 48 not upgraded.
Need to get 169 kB of archives.
After this operation, 869 kB of additional disk space will be used.
Get:1 http://ht.archive.ubuntu.com/ubuntu noble/main amd64 ufw all 0.36.2-6 [169 kB]
Fetched 169 kB in 9s (19.1 kB/s)
Preconfiguring packages ...
Selecting previously unselected package ufw.
(Reading database ... 135224 files and directories currently installed.)
Preparing to unpack .../archives/ufw_0.36.2-6_all.deb ...
Unpacking ufw (0.36.2-6) ...
Setting up ufw (0.36.2-6) ...
Processing triggers for rsyslog (8.2312.0-3ubuntu9.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
lens@lens:~$
```

Autorisation des ports

```
(be sure to update your rules accordingly)
lens@lens:~$ sudo ufw allow 2222/tcp
Skipping adding existing rule
Skipping adding existing rule (v6)
lens@lens:~$ sudo ufw allow 80/tcp
Rule added
Rule added (v6)
lens@lens:~$ sudo ufw allow 443/tcp
Rule added
Rule added (v6)
lens@lens:~$
```

Activation et verification de firewall

```

lens@lens:~$ sudo ufw enable
Firewall is active and enabled on system startup
lens@lens:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To Action From
--
22 ALLOW IN Anywhere
80 ALLOW IN Anywhere
443 ALLOW IN Anywhere
2222/tcp ALLOW IN Anywhere
80/tcp ALLOW IN Anywhere
443/tcp ALLOW IN Anywhere
22 (v6) ALLOW IN Anywhere (v6)
80 (v6) ALLOW IN Anywhere (v6)
443 (v6) ALLOW IN Anywhere (v6)
2222/tcp (v6) ALLOW IN Anywhere (v6)
80/tcp (v6) ALLOW IN Anywhere (v6)
443/tcp (v6) ALLOW IN Anywhere (v6)

lens@lens:~$

```

Protection contre brute force

Installation de fail2ban

```

lens@lens:~$ sudo apt install fail2ban -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
fail2ban is already the newest version (1.0.2-3ubuntu0.1).
0 upgraded, 0 newly installed, 0 to remove and 48 not upgraded.
lens@lens:~$

```

Gérer la configuration

```
GNU nano 7.2
[sshd]
enable = true
port = 2222
maxretry = 3
bantime = 600
```

Redemarrer et verifier le statut de fail2ban

```
lens@lens:~$ sudo systemctl restart fail2ban
lens@lens:~$ sudo fail2ban-client status
Status
|- Number of jail:      1
|- Jail list:   sshd
lens@lens:~$
```

Analyse des logs

Afficher les logs SSH et filtrer les erreurs


```

026-04-05T19:33:11.071466+00:00 lens sudo: lens : TTY=tty1 ; PWD=/home/lens ; USER=root ; COMMAND=/usr/bin/apt inst
026-04-05T19:33:11.073300+00:00 lens sudo: pam_unix(sudo:session): session opened for user root(uid=0) by lens(uid=1000)
026-04-05T19:33:11.655736+00:00 lens sudo: pam_unix(sudo:session): session closed for user root
026-04-05T19:33:38.864585+00:00 lens sudo: lens : TTY=tty1 ; PWD=/home/lens ; USER=root ; COMMAND=/usr/bin/apt inst
026-04-05T19:33:38.866726+00:00 lens sudo: pam_unix(sudo:session): session opened for user root(uid=0) by lens(uid=1000)
026-04-05T19:33:55.068288+00:00 lens sudo: pam_unix(sudo:session): session closed for user root
026-04-05T19:35:01.525067+00:00 lens CRON[2752]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
026-04-05T19:35:01.536189+00:00 lens CRON[2752]: pam_unix(cron:session): session closed for user root
026-04-05T19:36:21.575420+00:00 lens sudo: lens : TTY=tty1 ; PWD=/home/lens ; USER=root ; COMMAND=/usr/sbin/ufw def
026-04-05T19:36:21.585572+00:00 lens sudo: pam_unix(sudo:session): session opened for user root(uid=0) by lens(uid=1000)
026-04-05T19:36:22.464360+00:00 lens sudo: pam_unix(sudo:session): session closed for user root
026-04-05T19:36:45.527829+00:00 lens sudo: lens : TTY=tty1 ; PWD=/home/lens ; USER=root ; COMMAND=/usr/sbin/ufw def
026-04-05T19:36:45.535250+00:00 lens sudo: pam_unix(sudo:session): session opened for user root(uid=0) by lens(uid=1000)
026-04-05T19:36:46.716393+00:00 lens sudo: pam_unix(sudo:session): session closed for user root
026-04-05T19:37:55.489509+00:00 lens sudo: lens : TTY=tty1 ; PWD=/home/lens ; USER=root ; COMMAND=/usr/sbin/ufw all
026-04-05T19:37:55.492647+00:00 lens sudo: pam_unix(sudo:session): session opened for user root(uid=0) by lens(uid=1000)
026-04-05T19:37:55.593636+00:00 lens sudo: pam_unix(sudo:session): session closed for user root
026-04-05T19:38:08.593015+00:00 lens sudo: lens : TTY=tty1 ; PWD=/home/lens ; USER=root ; COMMAND=/usr/sbin/ufw all
026-04-05T19:38:08.595281+00:00 lens sudo: pam_unix(sudo:session): session opened for user root(uid=0) by lens(uid=1000)
026-04-05T19:38:08.742953+00:00 lens sudo: pam_unix(sudo:session): session closed for user root
026-04-05T19:38:18.290900+00:00 lens sudo: lens : TTY=tty1 ; PWD=/home/lens ; USER=root ; COMMAND=/usr/sbin/ufw all
026-04-05T19:38:18.292898+00:00 lens sudo: pam_unix(sudo:session): session opened for user root(uid=0) by lens(uid=1000)
026-04-05T19:38:18.437390+00:00 lens sudo: pam_unix(sudo:session): session closed for user root
026-04-05T19:39:01.542699+00:00 lens CRON[3597]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
026-04-05T19:39:01.549459+00:00 lens CRON[3597]: pam_unix(cron:session): session closed for user root
026-04-05T19:39:43.333344+00:00 lens sudo: lens : TTY=tty1 ; PWD=/home/lens ; USER=root ; COMMAND=/usr/sbin/ufw ena
026-04-05T19:39:43.340004+00:00 lens sudo: pam_unix(sudo:session): session opened for user root(uid=0) by lens(uid=1000)
026-04-05T19:39:43.642388+00:00 lens sudo: pam_unix(sudo:session): session closed for user root
026-04-05T19:40:02.764851+00:00 lens sudo: lens : TTY=tty1 ; PWD=/home/lens ; USER=root ; COMMAND=/usr/sbin/ufw sta
026-04-05T19:40:02.775875+00:00 lens sudo: pam_unix(sudo:session): session opened for user root(uid=0) by lens(uid=1000)
026-04-05T19:40:02.875052+00:00 lens sudo: pam_unix(sudo:session): session closed for user root
026-04-05T19:42:31.015908+00:00 lens sudo: lens : TTY=tty1 ; PWD=/home/lens ; USER=root ; COMMAND=/usr/bin/apt inst
026-04-05T19:42:31.018304+00:00 lens sudo: pam_unix(sudo:session): session opened for user root(uid=0) by lens(uid=1000)
026-04-05T19:42:31.997247+00:00 lens sudo: pam_unix(sudo:session): session closed for user root
026-04-05T19:44:07.095047+00:00 lens sudo: lens : TTY=tty1 ; PWD=/home/lens ; USER=root ; COMMAND=/usr/bin/nano /et
026-04-05T19:44:07.096887+00:00 lens sudo: pam_unix(sudo:session): session opened for user root(uid=0) by lens(uid=1000)
026-04-05T19:45:01.556070+00:00 lens CRON[3805]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
026-04-05T19:45:01.562066+00:00 lens CRON[3805]: pam_unix(cron:session): session closed for user root
026-04-05T19:45:33.694137+00:00 lens sudo: pam_unix(sudo:session): session closed for user root
026-04-05T19:46:48.797369+00:00 lens sudo: lens : TTY=tty1 ; PWD=/home/lens ; USER=root ; COMMAND=/usr/bin/systemct
026-04-05T19:46:48.800572+00:00 lens sudo: pam_unix(sudo:session): session opened for user root(uid=0) by lens(uid=1000)
026-04-05T19:46:49.323002+00:00 lens sudo: pam_unix(sudo:session): session closed for user root
026-04-05T19:47:12.482785+00:00 lens sudo: lens : TTY=tty1 ; PWD=/home/lens ; USER=root ; COMMAND=/usr/bin/fail2bar
026-04-05T19:47:12.485116+00:00 lens sudo: pam_unix(sudo:session): session opened for user root(uid=0) by lens(uid=1000)
026-04-05T19:47:12.567062+00:00 lens sudo: pam_unix(sudo:session): session closed for user root
026-04-05T19:48:02.091581+00:00 lens sudo: lens : TTY=tty1 ; PWD=/home/lens ; USER=root ; COMMAND=/usr/bin/apt inst
026-04-05T19:48:02.093567+00:00 lens sudo: pam_unix(sudo:session): session opened for user root(uid=0) by lens(uid=1000)
026-04-05T19:48:03.024542+00:00 lens sudo: pam_unix(sudo:session): session closed for user root
lens@lens:~$ grep "Failed password" /var/log/auth.log
lens@lens:~$

```

Afficher les logs Fail2ban

```
Lens [Running] - Oracle VirtualBox : 1
File Machine View Input Devices Help

2026-04-05 16:38:58,834 fail2ban.jail [666]: INFO Creating new jail 'sshd'
2026-04-05 16:39:00,997 fail2ban.jail [666]: INFO Jail 'sshd' uses systemd {}
2026-04-05 16:39:01,592 fail2ban.jail [666]: INFO Initiated 'systemd' backend
2026-04-05 16:39:01,623 fail2ban.filter [666]: INFO maxLines: 1
2026-04-05 16:39:01,647 fail2ban.filterssystemd [666]: INFO [sshd] Added journal match for: '_SYSTEMD_UNIT=sshd.servi
2026-04-05 16:39:01,647 fail2ban.filter [666]: INFO maxRetry: 5
2026-04-05 16:39:01,647 fail2ban.filter [666]: INFO findtime: 600
2026-04-05 16:39:01,647 fail2ban.actions [666]: INFO banTime: 600
2026-04-05 16:39:01,647 fail2ban.filter [666]: INFO encoding: UTF-8
2026-04-05 16:39:01,732 fail2ban.jail [666]: INFO Jail 'sshd' started
2026-04-05 16:39:02,839 fail2ban.filterssystemd [666]: INFO [sshd] Jail is in operation now (process new journal entr
2026-04-05 18:32:20,964 fail2ban.server [667]: INFO -----
2026-04-05 18:32:20,986 fail2ban.server [667]: INFO Starting Fail2ban v1.0.2
2026-04-05 18:32:20,991 fail2ban.observer [667]: INFO Observer start...
2026-04-05 18:32:21,121 fail2ban.database [667]: INFO Connected to fail2ban persistent database '/var/lib/fail2
2026-04-05 18:32:21,155 fail2ban.jail [667]: INFO Creating new jail 'sshd'
2026-04-05 18:32:23,439 fail2ban.jail [667]: INFO Jail 'sshd' uses systemd {}
2026-04-05 18:32:23,800 fail2ban.jail [667]: INFO Initiated 'systemd' backend
2026-04-05 18:32:23,811 fail2ban.filter [667]: INFO maxLines: 1
2026-04-05 18:32:23,819 fail2ban.filterssystemd [667]: INFO [sshd] Added journal match for: '_SYSTEMD_UNIT=sshd.servi
2026-04-05 18:32:23,819 fail2ban.filter [667]: INFO maxRetry: 5
2026-04-05 18:32:23,819 fail2ban.filter [667]: INFO findtime: 600
2026-04-05 18:32:23,819 fail2ban.actions [667]: INFO banTime: 600
2026-04-05 18:32:23,819 fail2ban.filter [667]: INFO encoding: UTF-8
2026-04-05 18:32:23,867 fail2ban.jail [667]: INFO Jail 'sshd' started
2026-04-05 18:32:24,827 fail2ban.filterssystemd [667]: INFO [sshd] Jail is in operation now (process new journal entr
2026-04-05 19:46:48,888 fail2ban.server [667]: INFO Shutdown in progress...
2026-04-05 19:46:48,890 fail2ban.observer [667]: INFO Observer stop ... try to end queue 5 seconds
2026-04-05 19:46:48,910 fail2ban.observer [667]: INFO Observer stopped, 0 events remaining.
2026-04-05 19:46:48,951 fail2ban.server [667]: INFO Stopping all jails
2026-04-05 19:46:49,275 fail2ban.actions [667]: NOTICE [sshd] Flush ticket(s) with nftables
2026-04-05 19:46:49,276 fail2ban.jail [667]: INFO Jail 'sshd' stopped
2026-04-05 19:46:49,276 fail2ban.database [667]: INFO Connection to database closed.
2026-04-05 19:46:49,276 fail2ban.server [667]: INFO Exiting Fail2ban
2026-04-05 19:46:49,413 fail2ban.server [3817]: INFO -----
2026-04-05 19:46:49,413 fail2ban.server [3817]: INFO Starting Fail2ban v1.0.2
2026-04-05 19:46:49,413 fail2ban.observer [3817]: INFO Observer start...
2026-04-05 19:46:49,415 fail2ban.database [3817]: INFO Connected to fail2ban persistent database '/var/lib/fail2
2026-04-05 19:46:49,416 fail2ban.jail [3817]: INFO Creating new jail 'sshd'
2026-04-05 19:46:49,494 fail2ban.jail [3817]: INFO Jail 'sshd' uses systemd {}
2026-04-05 19:46:49,497 fail2ban.jail [3817]: INFO Initiated 'systemd' backend
2026-04-05 19:46:49,497 fail2ban.filter [3817]: INFO maxLines: 1
2026-04-05 19:46:49,508 fail2ban.filterssystemd [3817]: INFO [sshd] Added journal match for: '_SYSTEMD_UNIT=sshd.servi
2026-04-05 19:46:49,508 fail2ban.filter [3817]: INFO maxRetry: 3
2026-04-05 19:46:49,508 fail2ban.filter [3817]: INFO findtime: 600
2026-04-05 19:46:49,508 fail2ban.actions [3817]: INFO banTime: 600
2026-04-05 19:46:49,508 fail2ban.filter [3817]: INFO encoding: UTF-8
2026-04-05 19:46:49,514 fail2ban.jail [3817]: INFO Jail 'sshd' started
2026-04-05 19:46:49,516 fail2ban.filterssystemd [3817]: INFO [sshd] Jail is in operation now (process new journal entr
lens@lens:~$
```

Test de sécurité

Scanner les ports du serveur avec :Nmap

```
lens@lens:~$ nmap 192.168.43.79
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-05 19:56 UTC
Nmap scan report for 192.168.43.79
Host is up (0.00044s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
443/tcp   closed https

Nmap done: 1 IP address (1 host up) scanned in 5.29 seconds
lens@lens:~$
```

Audit du serveur avec Lynis

Installation de lynis

```
lens@lens:~$ sudo apt install lynis -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  menu
Suggested packages:
  apt-listbugs debsecan debsums tripwire samhain aide menu-l10n gksu | kde-cli-tools | ktsuss
The following NEW packages will be installed:
  lynis menu
0 upgraded, 2 newly installed, 0 to remove and 48 not upgraded.
Need to get 602 kB of archives.
After this operation, 3,202 kB of additional disk space will be used.
Get:1 http://ht.archive.ubuntu.com/ubuntu noble/universe amd64 lynis all 3.0.9-1 [226 kB]
Get:2 http://ht.archive.ubuntu.com/ubuntu noble/universe amd64 menu amd64 2.1.50 [377 kB]
Fetched 602 kB in 10s (58.2 kB/s)
Selecting previously unselected package lynis.
(Reading database ... 135319 files and directories currently installed.)
Preparing to unpack .../archives/lynis_3.0.9-1_all.deb ...
Unpacking lynis (3.0.9-1) ...
Selecting previously unselected package menu.
Preparing to unpack .../archives/menu_2.1.50_amd64.deb ...
Unpacking menu (2.1.50) ...
Setting up lynis (3.0.9-1) ...
Created symlink /etc/systemd/system/timers.target.wants/lynis.timer → /usr/lib/systemd/system/lynis.timer.
lynis.service is a disabled or a static unit, not starting it.
Setting up menu (2.1.50) ...
Processing triggers for install-info (7.1-3build2) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for menu (2.1.50) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
lens@lens:~$
```

Vérification de l'installation

```
No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
lens@lens:~$ lynis --version
3.0.9
lens@lens:~$
```

Lancement d'un audit complet

```
2007-2021, CISofy - https://cisofy.com/lynis/
Enterprise support available (compliance, plugins, interface and tools)
#####

[+] Initializing program
-----
- Detecting OS... [ DONE ]
- Checking profiles... [ DONE ]

-----
Program version: 3.0.9
Operating system: Linux
Operating system name: Ubuntu
Operating system version: 24.04
Kernel version: 6.8.0
Hardware platform: x86_64
Hostname: lens
-----
Profiles: /etc/lynis/default.prf
Log file: /var/log/lynis.log
Report file: /var/log/lynis-report.dat
Report version: 1.0
Plugin directory: /etc/lynis/plugins
-----
Auditor: [Not Specified]
Language: en
Test category: all
Test group: all
-----
- Program update status... [ NO UPDATE ]

[+] System tools
-----
- Scanning available tools...
- Checking system binaries...

[+] Plugins (phase 1)
-----
Note: plugins have more extensive tests and may take several minutes to complete

- Plugin: debian
  [
[+] Debian Tests
-----
- Checking for system binaries that are required by Debian Tests...
- Checking /bin... [ FOUND ]
- Checking /sbin... [ FOUND ]
- Checking /usr/bin... [ FOUND ]
```



```

- Permissions for: /etc/sudoers [ OK ]
- Permissions for: /etc/sudoers.d/README [ OK ]
- PAM password strength tools [ SUGGESTION ]
- PAM configuration files (pam.conf) [ FOUND ]
- PAM configuration files (pam.d) [ FOUND ]
- PAM modules [ FOUND ]
- LDAP module in PAM [ NOT FOUND ]
- Accounts without expire date [ SUGGESTION ]
- Accounts without password [ OK ]
- Locked accounts [ OK ]
- Checking user password aging (minimum) [ DISABLED ]
- User password aging (maximum) [ DISABLED ]
- Checking expired passwords [ OK ]
- Checking Linux single user mode authentication [ OK ]
- Determining default umask
  - umask (/etc/profile) [ NOT FOUND ]
  - umask (/etc/login.defs) [ SUGGESTION ]
- LDAP authentication support [ NOT ENABLED ]
- Logging failed login attempts [ ENABLED ]

[+] Shells
-----
- Checking shells from /etc/shells
  Result: found 9 shells (valid shells: 9).
- Session timeout settings/tools [ NONE ]
- Checking default umask values
  - Checking default umask in /etc/bash.bashrc [ NONE ]
  - Checking default umask in /etc/profile [ NONE ]

[+] File systems
-----
- Checking mount points
  - Checking /home mount point [ SUGGESTION ]
  - Checking /tmp mount point [ SUGGESTION ]
  - Checking /var mount point [ SUGGESTION ]
- Checking LVM volume groups [ FOUND ]
  - Checking LVM volumes [ FOUND ]
- Query swap partitions (fstab) [ OK ]
- Testing swap partitions [ OK ]
- Testing /proc mount (hidepid) [ SUGGESTION ]
- Checking for old files in /tmp [ OK ]
- Checking /tmp sticky bit [ OK ]
- Checking /var/tmp sticky bit [ OK ]
- ACL support root file system [ ENABLED ]
- Mount options of / [ OK ]
- Mount options of /boot [ DEFAULT ]
- Mount options of /dev [ PARTIALLY HARDENED ]
- Mount options of /dev/shm [ PARTIALLY HARDENED ]
- Mount options of /run [ HARDENED ]

```

```
* Harden the system by installing at least one malware scanner, to perform periodic file system scans [HRDN-7230]
- Solution : Install a tool like rkhunter, chkrootkit, OSSEC
https://cisofy.com/lynis/controls/HRDN-7230/
```

Follow-up:

- Show details of a test (lynis show details TEST-ID)
- Check the logfile for all details (less /var/log/lynis.log)
- Read security controls texts (https://cisofy.com)
- Use --upload to upload data to central system (Lynis Enterprise users)

Lynis security scan details:

```
Hardening index : 63 [#####          ]
Tests performed : 282
Plugins enabled : 1
```

Components:

- Firewall [V]
- Malware scanner [X]

Scan mode:

```
Normal [V] Forensics [ ] Integration [ ] Pentest [ ]
```

Lynis modules:

- Compliance status [?]
- Security audit [V]
- Vulnerability scan [V]

Files:

- Test and debug information : /var/log/lynis.log
- Report data : /var/log/lynis-report.dat

Lynis 3.0.9

Auditing, system hardening, and compliance for UNIX-based systems
(Linux, macOS, BSD, and others)

2007-2021, CISOfy - https://cisofy.com/lynis/
Enterprise support available (compliance, plugins, interface and tools)

[TIP]: Enhance Lynis audits by adding your settings to custom.conf (see /etc/lynis/default.conf for all settings)

lens@lens:~\$

Analyse du rapport

```
GNU nano 7.2 /var/log/lynis.log
2026-04-05 20:05:05 Starting Lynis 3.0.9 with PID 4263, build date 2023-08-03
2026-04-05 20:05:05 ====
2026-04-05 20:05:05 ### 2007-2021, CISofy - https://ciscofy.com/lynis/ ###
2026-04-05 20:05:05 Checking permissions of /usr/share/lynis/include/profiles
2026-04-05 20:05:05 File permissions are OK
2026-04-05 20:05:05 Reading profile/configuration /etc/lynis/default.prfl
2026-04-05 20:05:05 Action: created temporary file /tmp/lynis.peg4ily3FR
2026-04-05 20:05:05 Language set via profile to ''
2026-04-05 20:05:05 Plugin 'authentication' enabled according profile (/etc/lynis/default.prfl)
2026-04-05 20:05:05 Plugin 'compliance' enabled according profile (/etc/lynis/default.prfl)
2026-04-05 20:05:05 Plugin 'configuration' enabled according profile (/etc/lynis/default.prfl)
2026-04-05 20:05:05 Plugin 'control-panels' enabled according profile (/etc/lynis/default.prfl)
2026-04-05 20:05:05 Plugin 'crypto' enabled according profile (/etc/lynis/default.prfl)
2026-04-05 20:05:05 Plugin 'dns' enabled according profile (/etc/lynis/default.prfl)
2026-04-05 20:05:05 Plugin 'docker' enabled according profile (/etc/lynis/default.prfl)
2026-04-05 20:05:05 Plugin 'file-integrity' enabled according profile (/etc/lynis/default.prfl)
2026-04-05 20:05:05 Plugin 'file-systems' enabled according profile (/etc/lynis/default.prfl)
2026-04-05 20:05:05 Plugin 'firewalls' enabled according profile (/etc/lynis/default.prfl)
2026-04-05 20:05:05 Plugin 'forensics' enabled according profile (/etc/lynis/default.prfl)
2026-04-05 20:05:05 Plugin 'hardware' enabled according profile (/etc/lynis/default.prfl)
2026-04-05 20:05:05 Plugin 'intrusion-detection' enabled according profile (/etc/lynis/default.prfl)
2026-04-05 20:05:05 Plugin 'intrusion-prevention' enabled according profile (/etc/lynis/default.prfl)
2026-04-05 20:05:05 Plugin 'kernel' enabled according profile (/etc/lynis/default.prfl)
2026-04-05 20:05:05 Plugin 'malware' enabled according profile (/etc/lynis/default.prfl)
2026-04-05 20:05:06 Plugin 'memory' enabled according profile (/etc/lynis/default.prfl)
2026-04-05 20:05:06 Plugin 'nginx' enabled according profile (/etc/lynis/default.prfl)
2026-04-05 20:05:06 Plugin 'pam' enabled according profile (/etc/lynis/default.prfl)
2026-04-05 20:05:06 Plugin 'processes' enabled according profile (/etc/lynis/default.prfl)
2026-04-05 20:05:06 Plugin 'security-modules' enabled according profile (/etc/lynis/default.prfl)
2026-04-05 20:05:06 Plugin 'software' enabled according profile (/etc/lynis/default.prfl)
2026-04-05 20:05:06 Plugin 'system-integrity' enabled according profile (/etc/lynis/default.prfl)
2026-04-05 20:05:06 Plugin 'systemd' enabled according profile (/etc/lynis/default.prfl)
2026-04-05 20:05:06 Plugin 'users' enabled according profile (/etc/lynis/default.prfl)
2026-04-05 20:05:06 Plugin 'debian' enabled according profile (/etc/lynis/default.prfl)
2026-04-05 20:05:07 Set option to default value: NTPD_ROLE --> client
2026-04-05 20:05:07 ====
2026-04-05 20:05:07 EOL check: 255
2026-04-05 20:05:07 Note: the end-of-life of 'Ubuntu 24.04.4 LTS' could not be checked. Entry missing in software-eol.db
2026-04-05 20:05:07 Program version: 3.0.9
2026-04-05 20:05:07 Operating system: Linux
2026-04-05 20:05:07 Operating system name: Ubuntu
2026-04-05 20:05:07 Operating system version: 24.04
2026-04-05 20:05:07 Kernel version: 6.8.0
2026-04-05 20:05:07 Kernel version (full): 6.8.0-106-generic
2026-04-05 20:05:07 Hardware platform: x86_64
2026-04-05 20:05:07 -----
[ Read 10300 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^J Execute    ^C Location   M-U Undo      M-A Set
^X Exit      ^R Read File  ^N Replace    ^U Paste      ^J Justify    ^_ Go To Line M-E Redo      M-G Copy
```

```
GNU nano 7.2 /var/log/lynis-report.dat
# Lynis Report
report_version_major=1
report_version_minor=0
report_datetime_start=2026-04-05 20:05:05
auditor=[Not Specified]
lynis_version=3.0.9
os=Linux
os_name=Ubuntu
os_fullname=Ubuntu 24.04.4 LTS
os_version=24.04
linux_version=Ubuntu
os_kernel_version=6.8.0
os_kernel_version_full=6.8.0-106-generic
hostname=lens
test_category=all
test_group=all
plugin_directory=/etc/lynis/plugins
lynis_update_available=0
suggestion[]=LYNIS|This release is more than 4 months old. Check the website or GitHub to see if there is an update available.
binaries_count=1727
binaries_suid_count=/usr/bin/chfn /usr/bin/chsh /usr/bin/fusermount /usr/bin/fusermount3 /usr/bin/gpasswd /usr/bin/mount
binaries_sgid_count=/usr/bin/chage /usr/bin/crontab /usr/bin/dotlockfile /usr/bin/expiry /usr/bin/ssh-agent /usr/sbin/passwd
binary_paths=/usr/bin,/usr/sbin,/usr/local/bin,/usr/local/sbin
vm=1
vmtype=virtualbox
container=0
systemd=1
plugin_enabled_phase1[]=debian[1.0.1]
slow_test[]=DEB-0001,14.482312
suggestion[]=DEB-0280|Install libpam-tmpdir to set $TMP and $TMPDIR for PAM sessions|-|-|
suggestion[]=DEB-0810|Install apt-listbugs to display a list of critical bugs prior to each APT installation|-|-|
suggestion[]=DEB-0811|Install apt-listchanges to display any significant changes prior to any upgrade via APT.|-|-|
plugins_enabled=1
hostid=e4b634f80fa1c43a4a3731c52da152aa0bc403ac
hostid2=0f97d8bea7bd805f7b95303311291a9e3e3b3b9c46e38398deecba60f623355
suggestion[]=B00T-5122|Set a password on GRUB boot loader to prevent altering boot configuration (e.g. boot in single user mode)
running_service_tool=systemctl
running_service[]=cron
running_service[]=dbus
running_service[]=fail2ban
running_service[]=fwupd
running_service[]=getty@tty1
running_service[]=ModemManager
running_service[]=multipathd
running_service[]=networkd-dispatcher
running_service[]=nginx

[ Read 778 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set
^X Exit      ^R Read File  ^N Replace    ^U Paste      ^J Justify    ^_ Go To Line  M-E Redo      M-G Cop
```

Affichage des suggestions avec grep


```

suggestion[] =PROC-3612|Check the output of ps for dead or zombie processes|-|-|
suggestion[] =AUTH-9229|Check PAM configuration, add rounds if applicable and expire passwords to encrypt with new values|-|-|
suggestion[] =AUTH-9230|Configure password hashing rounds in /etc/login.defs|-|-|
suggestion[] =AUTH-9262|Install a PAM module for password strength testing like pam_cracklib or pam_passwdqc|-|-|
suggestion[] =AUTH-9282|When possible set expire dates for all password protected accounts|-|-|
suggestion[] =AUTH-9286|Configure minimum password age in /etc/login.defs|-|-|
suggestion[] =AUTH-9286|Configure maximum password age in /etc/login.defs|-|-|
suggestion[] =AUTH-9328|Default umask in /etc/login.defs could be more strict like 027|-|-|
suggestion[] =FILE-6310|To decrease the impact of a full /home file system, place /home on a separate partition|-|-|
suggestion[] =FILE-6310|To decrease the impact of a full /tmp file system, place /tmp on a separate partition|-|-|
suggestion[] =FILE-6310|To decrease the impact of a full /var file system, place /var on a separate partition|-|-|
suggestion[] =USB-1000|Disable drivers like USB storage when not used, to prevent unauthorized storage or data theft|-|-|
suggestion[] =NAME-4028|Check DNS configuration for the dns domain name|-|-|
suggestion[] =PKGS-7370|Install debsums utility for the verification of packages with known good database|-|-|
suggestion[] =PKGS-7392|Update your system with apt-get update, apt-get upgrade, apt-get dist-upgrade and/or unattended-upgrades|-|-|
suggestion[] =PKGS-7394|Install package apt-show-versions for patch management purposes|-|-|
suggestion[] =NETW-3200|Determine if protocol 'dccp' is really needed on this system|-|-|
suggestion[] =NETW-3200|Determine if protocol 'sctp' is really needed on this system|-|-|
suggestion[] =NETW-3200|Determine if protocol 'rds' is really needed on this system|-|-|
suggestion[] =NETW-3200|Determine if protocol 'tipc' is really needed on this system|-|-|
suggestion[] =MAIL-8818|You are advised to hide the mail_name (option: smtpd_banner) from your postfix configuration. Use 'hide_mail_name' in /etc/postfix/main.cf|-|-|
suggestion[] =MAIL-8820:disable_vrfy_command|Disable the 'VRFY' command|disable_vrfy_command=no|text:run postconf -e disable_vrfy_command
|
suggestion[] =FIRE-4513|Check iptables rules to see which rules are currently not used|-|-|
suggestion[] =HTTP-6640|Install Apache mod_evasive to guard webserver against DoS/brute force attempts|-|-|
suggestion[] =HTTP-6643|Install Apache mod_security to guard webserver against web application attacks|-|-|
suggestion[] =HTTP-6710|Add HTTPS to nginx virtual hosts for enhanced protection of sensitive data and privacy|-|-|
suggestion[] =SSH-7408|Consider hardening SSH configuration|AllowTcpForwarding (set YES to NO)|-|-|
suggestion[] =SSH-7408|Consider hardening SSH configuration|ClientAliveCountMax (set 3 to 2)|-|-|
suggestion[] =SSH-7408|Consider hardening SSH configuration|LogLevel (set INFO to VERBOSE)|-|-|
suggestion[] =SSH-7408|Consider hardening SSH configuration|MaxSessions (set 10 to 2)|-|-|
suggestion[] =SSH-7408|Consider hardening SSH configuration|TCPKeepAlive (set YES to NO)|-|-|
suggestion[] =SSH-7408|Consider hardening SSH configuration|X11Forwarding (set YES to NO)|-|-|
suggestion[] =SSH-7408|Consider hardening SSH configuration|AllowAgentForwarding (set YES to NO)|-|-|
suggestion[] =LOGG-2154|Enable logging to an external logging host for archiving purposes and additional protection|-|-|
suggestion[] =LOGG-2190|Check what deleted files are still in use and why.|-|-|
suggestion[] =BANN-7126|Add a legal banner to /etc/issue, to warn unauthorized users|-|-|
suggestion[] =BANN-7130|Add legal banner to /etc/issue.net, to warn unauthorized users|-|-|
suggestion[] =ACCT-9622|Enable process accounting|-|-|
suggestion[] =ACCT-9626|Enable sysstat to collect accounting (disabled)|-|-|
suggestion[] =ACCT-9628|Enable auditd to collect audit information|-|-|
suggestion[] =FINT-4350|Install a file integrity tool to monitor changes to critical and sensitive files|-|-|
suggestion[] =TOOL-5002|Determine if automation tools are present for system management|-|-|
suggestion[] =FILE-7524|Consider restricting file permissions|See screen output or log file|text:Use chmod to change file permissions|-|-|
suggestion[] =KRNL-6000|One or more sysctl values differ from the scan profile and could be tweaked||Change sysctl value (sysctl-key)|
suggestion[] =HRDN-7230|Harden the system by installing at least one malware scanner, to perform periodic file system scans|-|-|
lens@lens:~$

```

Amélioration de la sécurité

Installation de ufw et aide

```

Creating config file /etc/aide/aide.conf.d/31_aide_spampd with new version
Creating config file /etc/aide/aide.conf.d/31_aide_webalizer with new version
Creating config file /etc/aide/aide.conf.d/31_aide_apt-listbugs with new version
Creating config file /etc/aide/aide.conf.d/31_aide_borgbackup with new version
Creating config file /etc/aide/aide.conf.d/31_aide_nrpe with new version
Creating config file /etc/aide/aide.conf.d/31_aide_trac with new version
Creating config file /etc/aide/aide.conf.d/31_aide_ddclient with new version
Creating config file /etc/aide/aide.conf.d/31_aide_lastlog with new version
Creating config file /etc/aide/aide.conf.d/31_aide_ssh-agent with new version
Creating config file /etc/aide/aide.settings.d/31_aide_trac_settings with new version
Creating config file /etc/aide/aide.settings.d/31_aide_svn-server_settings with new version
Creating config file /etc/aide/aide.settings.d/31_aide_torrus_settings with new version
Creating config file /etc/aide/aide.settings.d/31_aide_apt_settings with new version
Creating config file /etc/aide/aide.settings.d/10_aide_sourceslist with new version
Creating config file /etc/cron.daily/dailyaidecheck with new version
Creating group '_aide' with GID 988.
Creating user '_aide' (Advanced Intrusion Detection Environment) with UID 988 and GID 988.
Created symlink /etc/systemd/system/timers.target.wants/dailyaidecheck.timer → /usr/lib/systemd/system/dailyaidecheck.timer.
dailyaidecheck.service is a disabled or a static unit, not starting it.
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
lens@lens:~$

```

Activation des mises à jour automatiques

```

lens@lens:~$ sudo apt install unattended-upgrades
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
unattended-upgrades is already the newest version (2.9.1+nmu4ubuntu1).
unattended-upgrades set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 48 not upgraded.
lens@lens:~$

```

Relancement de l'audit

```

* Harden the system by installing at least one malware scanner, to perform periodic file system scans [HRDN-7230]
- Solution : Install a tool like rkhunter, chkrootkit, OSSEC
  https://cisofy.com/lynis/controls/HRDN-7230/

Follow-up:
-----
- Show details of a test (lynis show details TEST-ID)
- Check the logfile for all details (less /var/log/lynis.log)
- Read security controls texts (https://cisofy.com)
- Use --upload to upload data to central system (Lynis Enterprise users)

=====

Lynis security scan details:

Hardening index : 64 [#####          ]
Tests performed : 285
Plugins enabled : 1

Components:
- Firewall          [V]
- Malware scanner   [X]

Scan mode:
Normal [V] Forensics [] Integration [] Pentest []

Lynis modules:
- Compliance status [?]
- Security audit    [V]
- Vulnerability scan [V]

Files:
- Test and debug information : /var/log/lynis.log
- Report data                : /var/log/lynis-report.dat

=====

Lynis 3.0.9

Auditing, system hardening, and compliance for UNIX-based systems
(Linux, macOS, BSD, and others)

2007-2021, CISOfy - https://cisofy.com/lynis/
Enterprise support available (compliance, plugins, interface and tools)

=====

[TIP]: Enhance Lynis audits by adding your settings to custom.prfl (see /etc/lynis/default.prfl for all settings)

lens@lens:~$

```

Amélioration de la sécurité du serveur

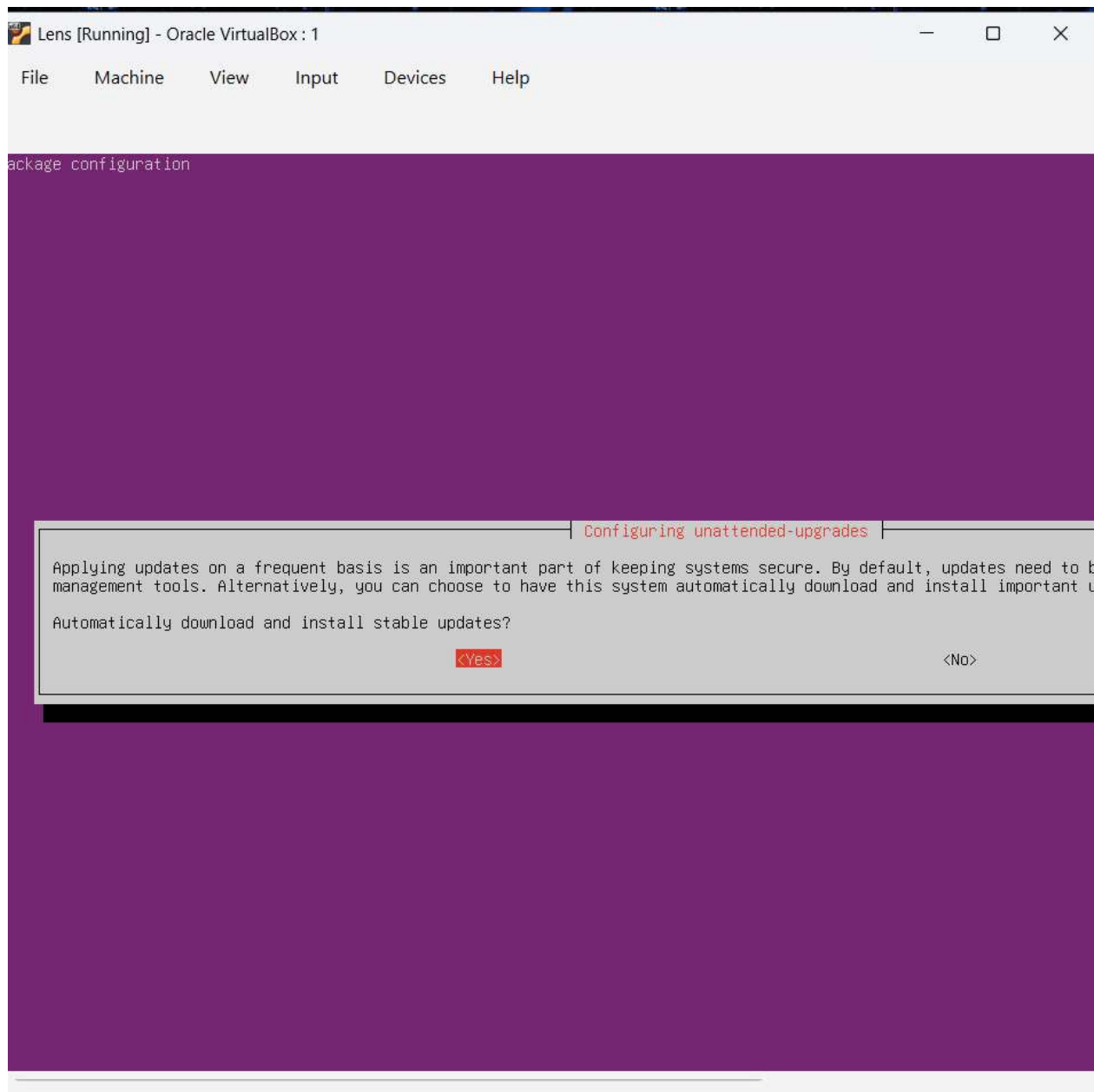
Installation du service

```

Apr 05 18:32:21 lens systemd[1]: Started unattended-upgrades.service - Unattended Upgrades Shutdown.
lens@lens:~$ sudo apt install unattended-upgrades -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
unattended-upgrades is already the newest version (2.9.1+nmu4ubuntu1).
0 upgraded, 0 newly installed, 0 to remove and 48 not upgraded.
lens@lens:~$

```

Configuration



Vérification de fonctionnement

```

lens@lens:~$ systemctl status unattended-upgrades
• unattended-upgrades.service - Unattended Upgrades Shutdown
   Loaded: loaded (/usr/lib/systemd/system/unattended-upgrades.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-04-05 18:32:21 UTC; 1h 53min ago
     Docs: man:unattended-upgrade(8)
    Main PID: 756 (unattended-upgr)
      Tasks: 2 (limit: 2263)
     Memory: 9.4M (peak: 9.7M)
        CPU: 144ms
    CGroup: /system.slice/unattended-upgrades.service
            └─756 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal

Apr 05 18:32:21 lens systemd[1]: Started unattended-upgrades.service - Unattended Upgrades Shutdown.
lens@lens:~$

```

Installer un système de détection d'intégrité

Installation de aide et initialization de BD

```

Apr 05 18:32:21 lens systemd[1]: Started unattended-upgrades.service - Unattended Upgrades Shutdown.
lens@lens:~$ sudo apt install unattended-upgrades -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
unattended-upgrades is already the newest version (2.9.1+nmu4ubuntu1).
0 upgraded, 0 newly installed, 0 to remove and 48 not upgraded.
lens@lens:~$ sudo apt install aide -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
aide is already the newest version (0.18.6-2ubuntu0.1).
0 upgraded, 0 newly installed, 0 to remove and 48 not upgraded.
lens@lens:~$ sudo aideinit
Running aide --init...
^X^Z
[4]+  Stopped                  sudo aideinit
lens@lens:~$ sudo aideinit
Overwrite existing /var/lib/aide/aide.db.new [Yn]? y
Running aide --init...
AIDE --init return code 21
lens@lens:~$

```

Vérification des permissions du fichier shadow

```

lens@lens:~$ ls -l /etc/shadow
-rw-r----- 1 root shadow 1305 Apr  5 20:15 /etc/shadow
lens@lens:~$

```

Vérifier les permissions du dossier SSH et correction

```
lens@lens:~$ ls -ld ~/.ssh
drwx----- 2 lens lens 4096 Mar 25 00:29 /home/lens/.ssh
lens@lens:~$ chmod 700 ~/.ssh
lens@lens:~$ chmod 600 ~/.ssh/authorized_keys
lens@lens:~$
```

Vérifier l'espace disque et les logs

Logrotate déjà installé

Relancement de l'audit de sécurité

```
* Harden the system by installing at least one malware scanner, to perform periodic file system scans [HRDN-7230]
- Solution : Install a tool like rkhunter, chkrootkit, OSSEC
https://cisofy.com/lynis/controls/HRDN-7230/

Follow-up:
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- Show details of a test (lynis show details TEST-ID)
- Check the logfile for all details (less /var/log/lynis.log)
- Read security controls texts (https://cisofy.com)
- Use --upload to upload data to central system (Lynis Enterprise users)

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Lynis security scan details:

Hardening index : 64 [#####          ]
Tests performed : 285
Plugins enabled : 1

Components:
- Firewall           [V]
- Malware scanner    [X]

Scan mode:
Normal [V]  Forensics []  Integration []  Pentest []

Lynis modules:
- Compliance status  [?]
- Security audit     [V]
- Vulnerability scan [V]

Files:
- Test and debug information : /var/log/lynis.log
- Report data                : /var/log/lynis-report.dat

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Lynis 3.0.9

Auditing, system hardening, and compliance for UNIX-based systems
(Linux, macOS, BSD, and others)

2007-2021, CISOfy - https://cisofy.com/lynis/
Enterprise support available (compliance, plugins, interface and tools)

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[TIP]: Enhance Lynis audits by adding your settings to custom.prf (see /etc/lynis/default.prf for all settings)

lens@lens:~$
```

Questions

1. Pourquoi les mises à jour automatiques sont-elles importantes en production ?

Les mises à jour automatiques permettent de corriger rapidement les failles de sécurité, bugs et vulnérabilités sans dépendre d'une intervention manuelle. En production, cela réduit le risque d'exploitation par des attaquants, assure la stabilité des services et garantit que le système bénéficie des dernières optimisations. Sans elles, un serveur peut rester exposé à des menaces connues.

2. Quel est le rôle d'un IDS comme AIDE ?

Un IDS (Intrusion Detection System) comme **AIDE (Advanced Intrusion Detection Environment)** surveille l'intégrité des fichiers système. Il crée une base de référence (hashes, permissions, métadonnées) et alerte lorsqu'une modification non autorisée est détectée. Son rôle est de repérer les intrusions ou altérations suspectes, par exemple un fichier système modifié par un malware ou un attaquant.

3. Pourquoi sécuriser les permissions des fichiers système ?

Les fichiers système contiennent des configurations critiques et des exécutables essentiels. Si leurs permissions sont trop larges, un utilisateur ou un processus malveillant peut les modifier, ce qui compromet la stabilité et la sécurité du serveur. En restreignant les droits (lecture, écriture, exécution), on limite les risques d'escalade de privilèges et de compromission du système.

4. Peut-on considérer un serveur totalement sécurisé ?

Non, un serveur ne peut jamais être considéré comme **totalement sécurisé**. La sécurité est un processus continu: de nouvelles vulnérabilités apparaissent régulièrement, et aucun système n'est invulnérable. On peut seulement réduire les risques au maximum en appliquant des bonnes pratiques (mises à jour, surveillance, durcissement, sauvegardes), mais il faut toujours rester vigilant et prêt à réagir.

Cocclusion.

Ce TD m'a permis de savoir comment sécuriser un serveur Linux, protéger les accès SSH, analyser la sécurité d'un système, interpréter un rapport d'audit, identifier les failles potentielles et proposer des améliorations de sécurité, donc la tâche a été réussie.